

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME:

Cloud Computing and
Big Data Analytics

COURSE CODE:

CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

ave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct

Sumudhara 192311332 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

Understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Sumudhara*

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services.
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to

Describe the evolution from standalone computing to cloud-based system

standalone computing: application and data were stored on a single computer with limited storage and processing power.

Client-server computing: multiple clients connected to centralized services for data sharing and resource management.

Distributed computing: workloads were shared across multiple computers improving performance and reliability.

Grid computing: combined resources from different locations to solve large computational problems.

Virtualization: Hypervisors enables multiple virtual machines (VMs) to run on a single physical server, improving hardware utilization.

Cloud computing: provides computing resources (storage, distributed networking, software) over the internet with on-demand access.

Interact working of hypervisors in virtualization. Difference b/w Type 1 and Type 2 hypervisors with example.

A hypervisor is software that creates and manages virtual machines. It allocates CPU, memory storage, and network resources to each while keeping them isolated.

Type 1 Hypervisors:-

- runs directly on hardware (bare metal)
- faster and more secure
- used in data centers and cloud.

eg: VMware ESXi, Microsoft Hyper-V, Xen.

Type 2 Hypervisors.

- runs on top of a host operating system
- easier to install but slightly slower.
- used for personal computers and testing

eg: Oracle VM Virtual Box

3. Describe the role of APIs in cloud service integration. & APIs support interoperability between services.

An API (Application Programming Interface) is a set of rules that different software applications use to communicate with each other. Computing APIs help connect cloud services applications.

Role of APIs:-

- Enable communication between cloud services
- Automate cloud operations
- Support data exchange between different platforms

→ Improve interoperability between applications

4) Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.

Resource pooling is a cloud computing feature where computing resources such as servers, storage and networks are shared among multiple users. Resources are allocated dynamically based on user demand.

Benefits :

- Efficient use of resources.
- Reduced operation cost.
- High availability.
- Scalability.

Interpret the concept of rapid elasticity in cloud. Explain how it supports dynamic workloads.

Rapid elasticity is the ability of cloud computing to ~~scale~~ increase or decrease computing resources on user demand. Resources are automatically scaled up during high workload and scaled down when decreases.

Benefits :-

- Handles sudden traffic increase
- Reduces costs by using cloud services
- Improves application performance
- Provides flexibility and scalability

Develop

↓
Create

break